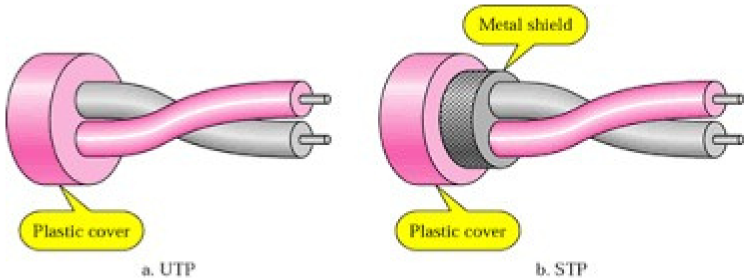
Lab - 1
Network Wiring and LAN Setup

Theory:

A. Ethernet

Ethernetis a family ofcomputer networkingtechnologies commonly used inlocal area networks (LANs) andmetropolitan area networks(MANs). **Ethernet over twisted pair** technologies usetwisted-pair cablesfor thephysical layerof an Ethernet computer network. Physical connection for networkingshowed the potential of simple *unshielded twisted pair*(UTP) for many applications in small networks like LAN and MAN with different category cables. Later the advancement in category of such UTP/STP cable leads to cat5/6 cables in use.



Internal Cable Structure

B.IP addresses

IP addresses are the Network ID given to network devices to identify them on the network and to enable different devices to interact with each other. Unlike the physical addresses (MAC address) which is a permanent address set by the manufacturer, the IP addresses are either configured manually (static IP address) or they are configured by a DHCP server, where the IP addresses assigned to the clients changes every time the device boots up.

C.Subnet Mask

A subnet mask basically gives information on network and host portion of the address. It also helps to identify which part of IP address is reserved for the network and which part is available for host use. In short, it also enables to calculate if two IP addresses are in the same subnet or not.

D.DNS – Domain Name System

This is a server that translates domain names into IP addresses. Because domain names are alphabetic, they're easier to remember. The Internet however, is really based on IP addresses. Every time you use a domain name, therefore, a DNS service must translate the name into the corresponding IP address.

E. Gateway

A **gateway** is a **network** point that acts as an entrance to another **network**. On the Internet, a node or stopping point can be either a **gateway** node or a host (end-point) node. Both the computers of Internet users and the computers that serve pages to users are host nodes.

F. Some useful network commands:(Explain yourself)

I. **ipconfig** – Quickly Find Your IP Address with additional commands like: /all, /flushdns, /release, /renew etc.

II. **ping**– Troubleshoot Network Connection Issues

Method:

A. Cable crimping

I. Standard, Straight-Through Wiring Diagram (both ends are the same)

RJ45 Pin #	Wire Color (T568B)
1	White/ Orange
2	Orange
3	White/ Green
4	Blue
5	White/Blue
6	Green
7	White/ Brown
8	Brown

Straight-Through Ethernet Cable Pin Out

I. Crossover Cable Wiring Diagram(T568B):

RJ45 Pin # (END 1)	Wire Color		RJ45 Pin # (END 2)	Wire Color	
1	White/ Orange		1	White/ Green	
2	Orange		2	Green	
3	White/ Green		3	White/ Orange	
4	Blue		4	White/ Brown	
5	White/Blue		5	Brown	
6	Green		6	Orange	
7	White/ Brown		7	Blue	
8	Brown		8	White/Blue	

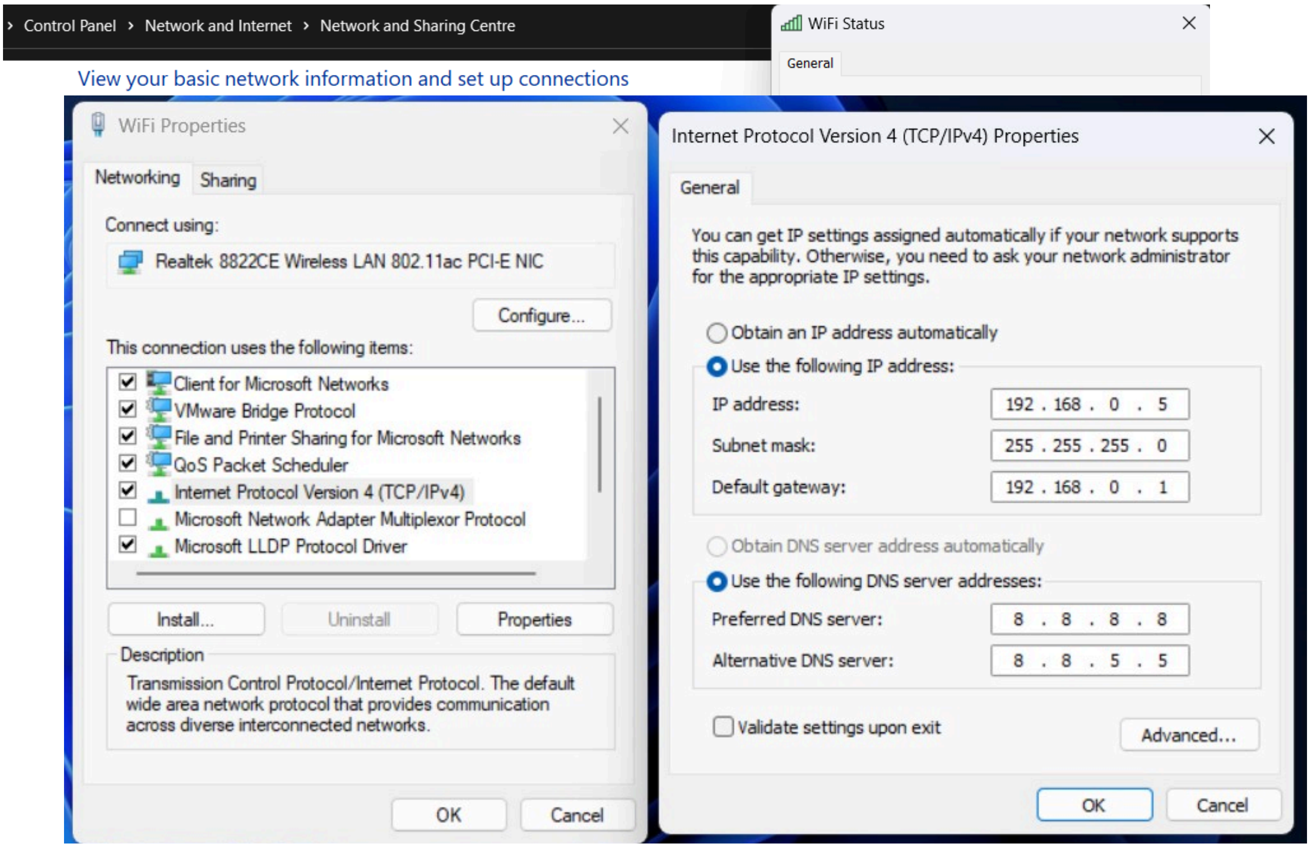
A.Network setting (windows)

I. Configuring network Manually

(use your own configuration configuration manually)

Steps :-

1)Open Control panel
2)Go to “Network and Internet”
3)Select “Network status” and select “Proporties”
4)Double click on “Internet Protocol Version 4 (TCP/IPv4)
5)Now update IP address, Subnet mask, Default gateway and DNS manually
6)Select Ok and exit



II. Screenshot for “ping” and “ipconfig” command. (use your own screen shot)

Conclusion:

From this practical we learned about some of important network components and learned manually configuration of Network practically.

Lab - 2
Networking Devices and Firewall

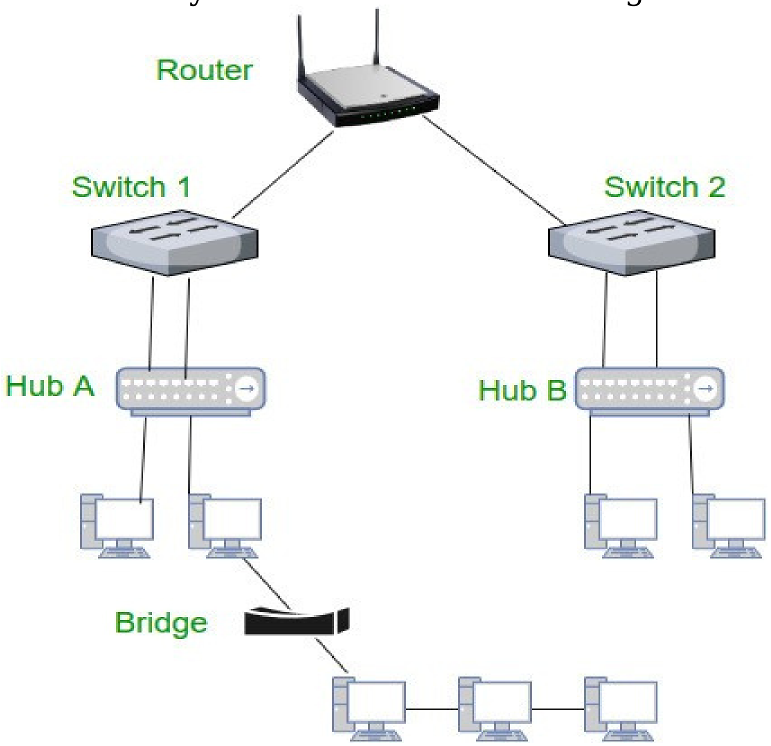
Theory:

A.Hub: A hub connects multiple wires coming from different branches, for example, the connector in star topology which connects different stations. Hubs cannot filter data, so data packets are sent to all connected devices.

B. Bridge: A bridge is a repeater with add on functionality of filtering content by reading the MAC addresses of source and destination. It has a single input and single output port, thus making it a 2-port device.

C. Switch: A switch is a multi-port bridge with a buffer and a design that can boost its efficiency and performance. Switch is a data link layer device. Switch can perform error checking before forwarding data that makes it very efficient as it does not forward packets that have errors and forward good packets selectively to correct port only.

D. Routers: A router is a device like a switch that routes data packets based on their IP addresses. Router is mainly a Network Layer device. Routers normally connect LANs and WANs together and have a dynamically updating routing table based on which they make decisions on routing the data packets.

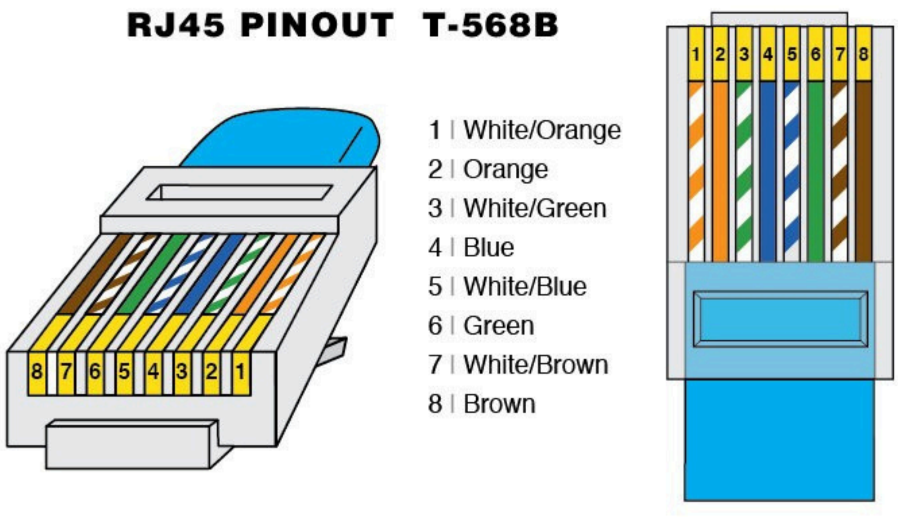


E. Gateway: A gateway, as the name suggests, is a passage to connect two networks together that may work upon different networking models. They basically work as the messenger agents that take data from one system, interpret it, and transfer it to another system.

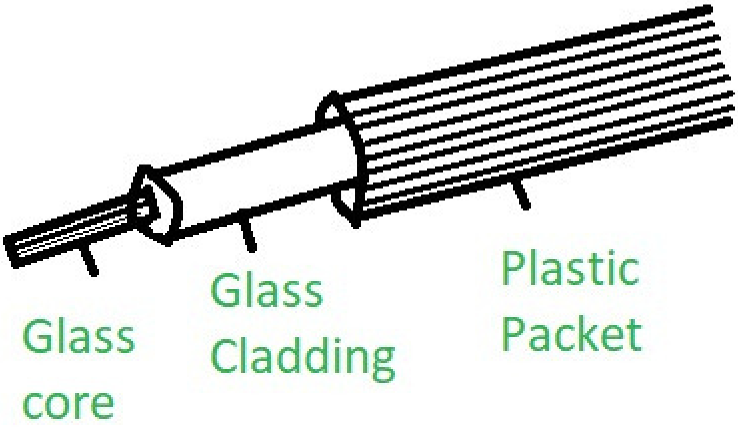
F.Explain about different media (Guided – cat6, fiber and coaxial cable -- Unguided Wi-Fi)

Guided media are also known as wired or bounded media. These media consist of wires through which the data is transferred. Guided media is a physical link between transmitter and recipient devices. Signals are directed in a narrow pathway using physical links. These media types are used for shorter distances since physical limitation limits the signal that flows through these transmission media.

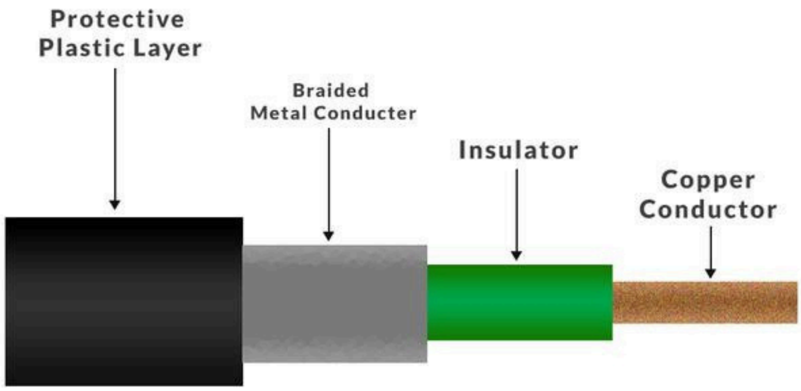
CAT6: - Category 6 is an Ethernet cable standard defined by the Electron Industries Association and Telecommunications Industry Association. Cat 6 is the sixth generation of twisted pair Ethernet cabling that is used in home and business networks.



Fiber cable: - Fiber-optic cable contains anywhere from a few to hundreds of optical fibers within a plastic casing. Also known as optic cables or optical fiber cables, they transfer data signals in the form of light and travel hundreds of miles significantly faster than those used in traditional electrical cables. And because fiber-optic cables are non-metallic, they are not affected by electromagnetic interference (i.e. lightning) that can reduce speed of transmission. Fiber cables are also safer as they do not carry a current and therefore cannot generate a spark.



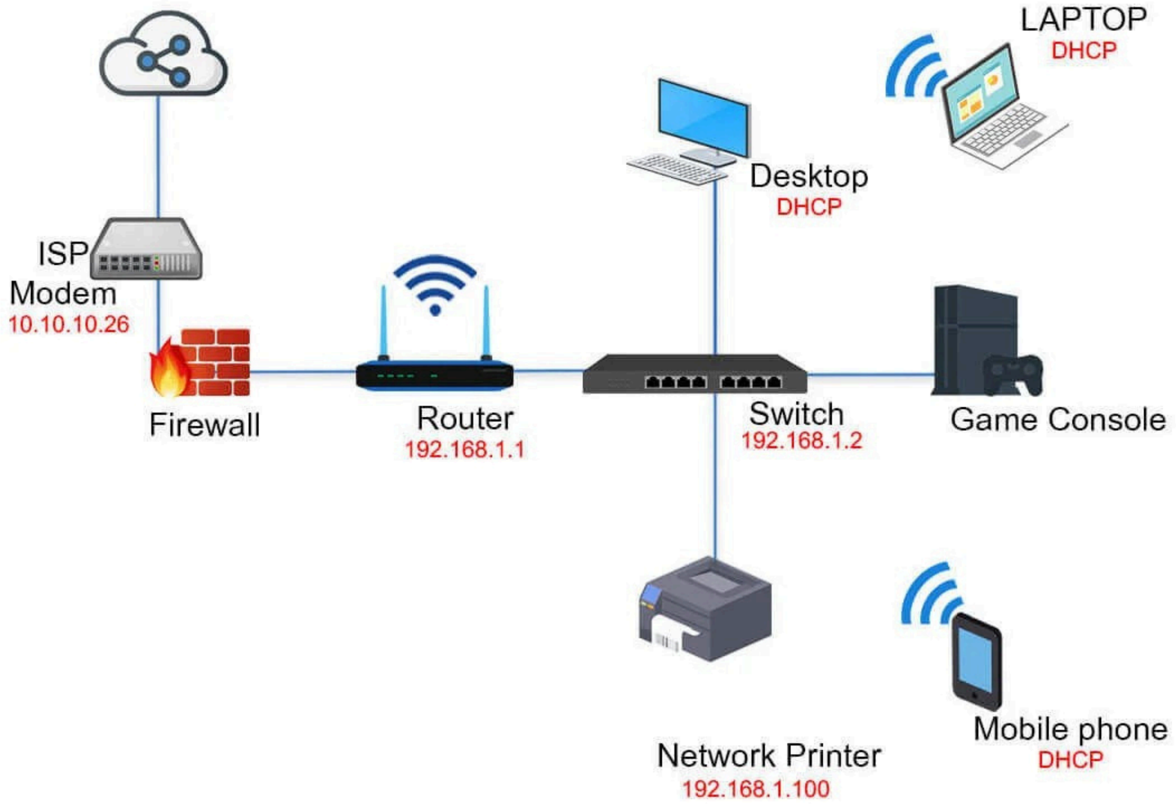
Coaxial Cable: - A coaxial cable is an electrical cable with a copper conductor
and an insulator shielding around it and a braided metal mesh that prevents signal interference and cross talk. Coaxial cable is also known as coax. The core copper conductor is used for the transmission of signals and the insulator is used to provide insulation to the copper conductor and the insulator is surrounded by a braided metal conductor which helps to prevent the interference of electrical signals and prevent cross talk. This entire setup is again covered with a protective plastic layer to provide extra safety to the cable.



Method:

1. **Include the picture of the devices.**
2. **Explain how internet comes in your home with network diagram. (Include type of link used.**

How does the internet come into your home?



Internet Service Provider (ISP): This is the company that provides internet connectivity to your home. ISPs offer various types of internet connections such as cable, DSL, fiber-optic, or satellite.
Modem: The modem is the device provided by the ISP or purchased separately. It's responsible for converting the incoming signal from the ISP into a form that your network devices can use. Modems come in different types depending on the type of internet connection, such as cable modems, DSL modems, or fiber-optic modems.
Router/Gateway: The router or gateway connects to the modem and serves as the central hub for your home network. It allows multiple devices to connect to the internet simultaneously and facilitates communication between devices within the network. Routers can be wired or wireless, and many modern routers often combine features of both. They typically have multiple Ethernet ports for wired connections and built-in Wi-Fi for wireless connections.
Wired Devices: These are devices such as desktop computers, gaming consoles, smart TVs, or streaming devices that connect to the router via Ethernet cables for a stable and high-speed connection.
Wireless Devices: These include smartphones, laptops, tablets, smart home devices, and other gadgets that connect to the router using Wi-Fi. The router broadcasts a wireless signal that these devices can detect and connect to, providing internet access without the need for physical cables.

Conclusion: -

Lab 3

Packet Tracer

1.Installation process of packet tracer (list the step).

Windows

a. Navigate to the location where you saved the Packet Tracer installation file. b. Double click the Packet Tracer installation file. c. The Setup dialog opens to the License Agreement. Read through the agreement,
click I accept the agreement, and then click Next.

d. In the Select Destination Location window, choose a location or accept the default location, and then click Next.

e. In the Select Start Menu Folder window, you can choose a different installation folder, if you prefer, or accept the default location, and then click Next.

f. In the Select Additional Tasks window, you can choose your Additional shortcuts, and then click Next.

g. In the Ready to Install window, review your installation settings. If you need to change anything, click Back to navigate back to a setting you wish to modify. When ready, click Install to begin the installation process.

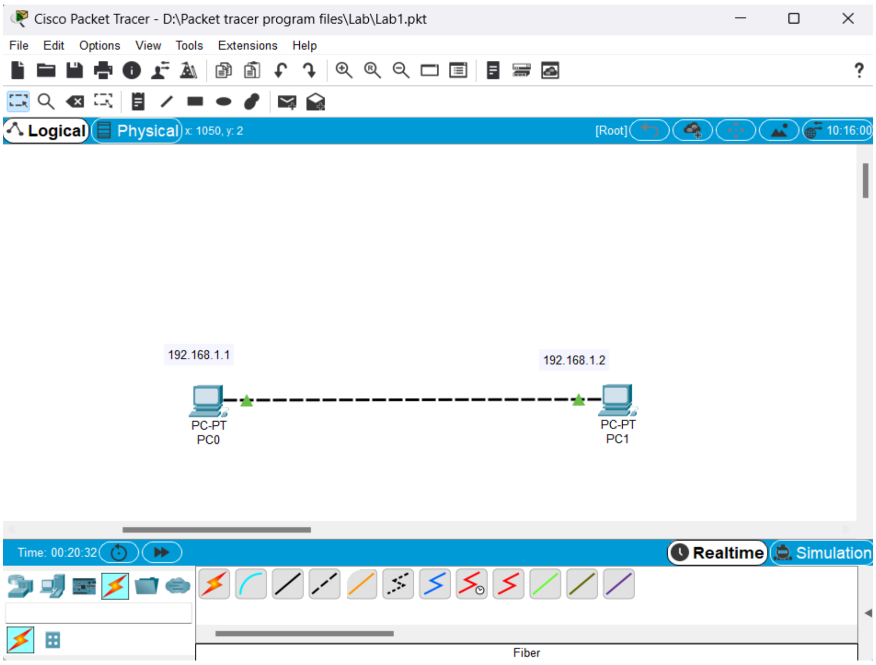
2.Connect two computers using cross cable in packet tracer. Verify the connection using Ping Command. (Show all the steps)

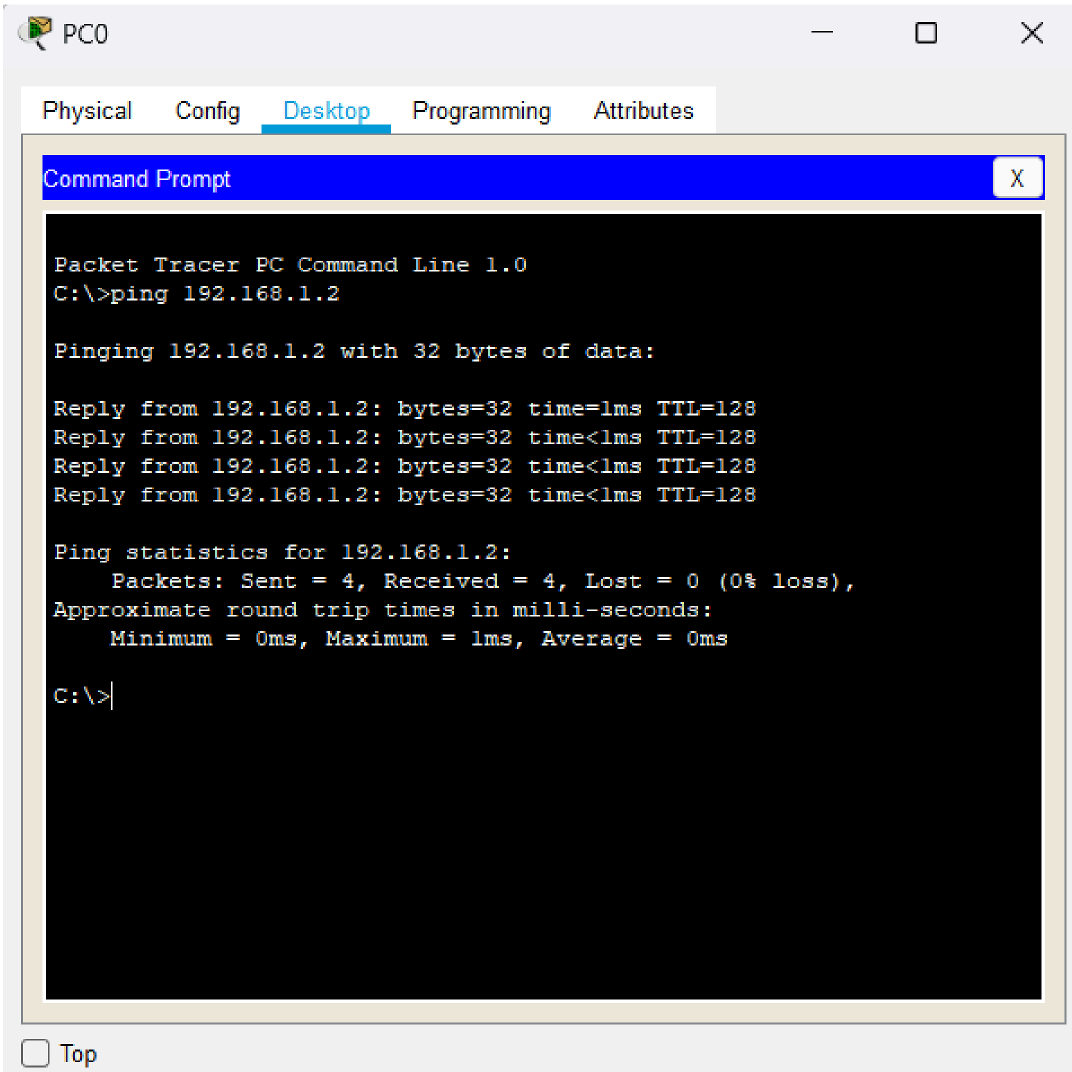
Step 1) Insert two PC from ‘End Devices’ menu into the drawing window of Packet Tracer.

Step 2) Select a copper cross over wire from ‘Connections’ menu and join the two PC as FastEthernet0.

Step 3) Configure both PC’s IP address by clicking to the PC symbol and selecting FastEthernet0 from Config menu and make sure to enable by clicking ‘On tick’ box.

Step 4) Go to PC0 and navigate to Desktop and then command prompt and type command as shown below.


Cisco Packet Tracer - D:\Packet tracer program files\Lab\Lab1.pkt
File Edit Options View Tools Extensions Help
Logical
Physical
x: 1050, y: 2
[Root]
10:16:00
192.168.1.1
192.168.1.2
PC-PT
PC0
PC-PT
PC1
Time: 00:20:32
Realtime
Simulation
Fiber


PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
Top

Lab 4

1. Explain modes of cisco router

There are mainly 5 modes in the router:

User execution mode –

As soon as the interface up message appears and press enter, the router> prompt will pop up. This is called user execution mode. This mode is limited to some monitoring commands.

Privileged mode –

As we type enable to user mode, we enter Privileged mode where we can view and change the configuration of the router. Different commands like show running configuration, show IP interface brief, etc can run on this mode which is used for troubleshooting purposes.

Global configuration mode –

As we type configure terminal to the user mode, we will enter the global configuration mode. Commands entered in these modes are called global commands and they affect the running configuration of the router. In this mode, a different configuration like making a local database on the router by providing username and password can set enable and secret password, etc.

Interface configuration mode –

In this mode, only the configuration of interfaces is done. Assigning an IP address to an interface, bringing up the interface are the common tasks done in this mode.

ROMMON mode –

We can enter this mode when we interrupt the boot process of the router. Generally, we enter in this mode during the password recovery process or backing up of IOS on devices like TFTP server. It is like the BIOS mode of a PC.

2.Basic router configuration command

Configuration –

The user execution mode:

```
router>
```

Entering into privilege mode from user execution mode:

```
router>enable
router#
```

Exiting from privilege mode to user execution mode:

```
router#disable
router>
```

Entering in global configuration mode from privilege mode:

```
router#configure terminal
router(config)#
```

Exiting from global configuration mode to privilege mode:

```
router(config)#exit
router#
```

Entering into interface mode from global configuration mode. Here we have to specify the router's interface.

```
router(config)#interface fa0/0
router(config-if)#
```

Exiting from interface mode to global configuration mode.

```
router(config-if)#exit
router(config)#
```

Exiting from interface mode to privilege mode.

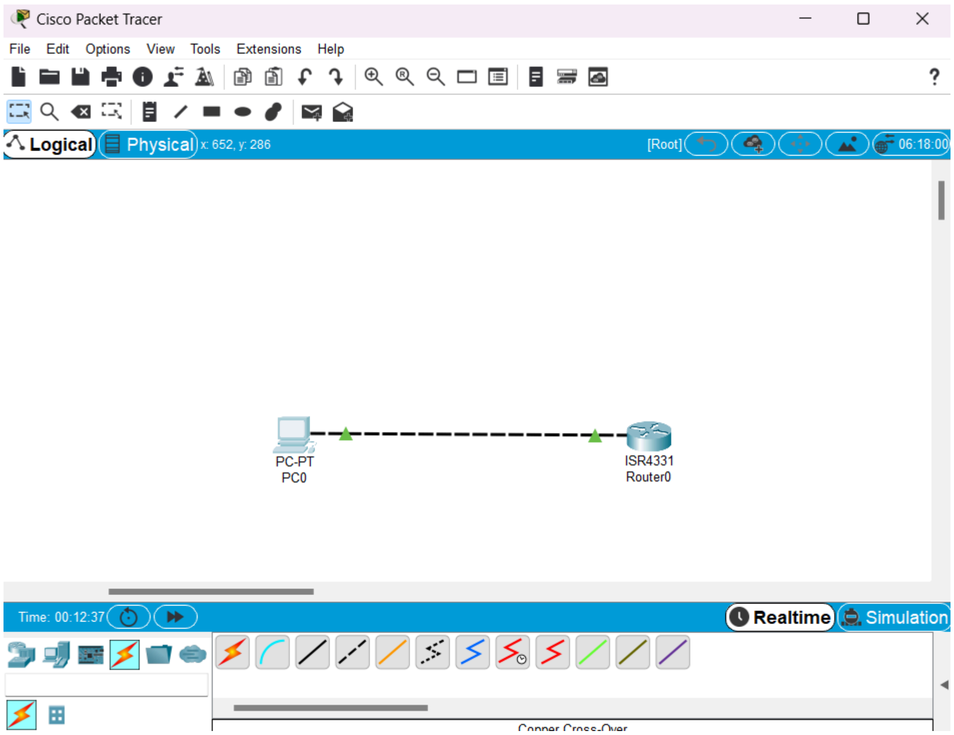
```
router(config-if)#end
router#
```

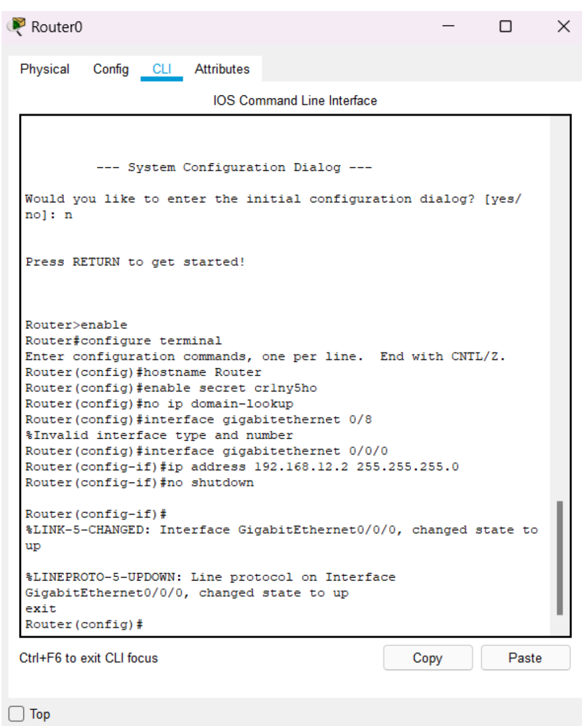
Lab 5

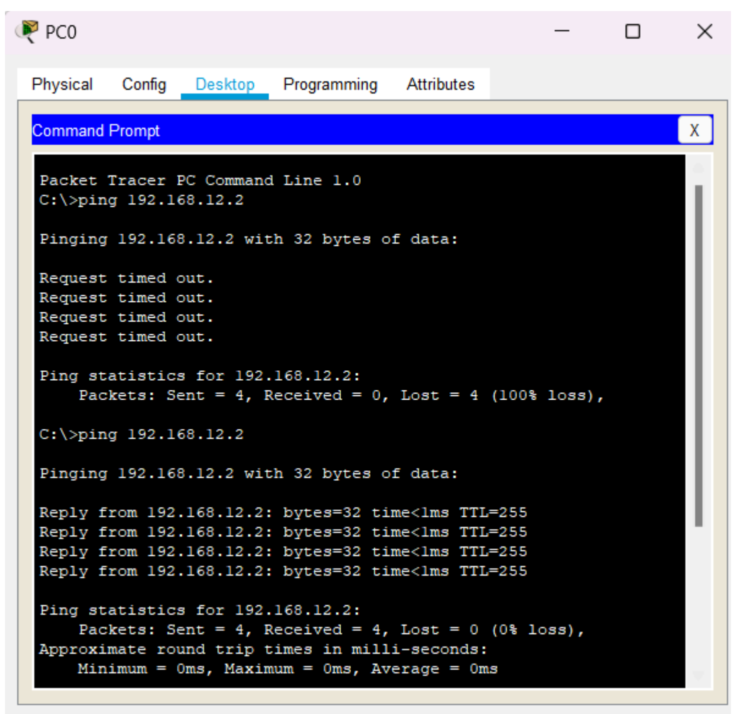
1.Connect a computer with router and verify the connection using Ping command. Include all the screenshots of the result.

Steps: -

a)Creating a connection between a router and a computer.
b)Configuring Ip address for the router and the computer.
c)Check the connection using ping command in the terminal.







LAB 6

Q. Connect two router and static configuration

I have configured static routing using two routers in just two simple steps:

First: Assign IP address on Routers

Second: Configure Static Routing on Both Routers.

First:

Assign IP address on Router R1:

Router(config)#host r1

r1(config)#int fa0/0

r1(config-if)#ip add 2.0.0.1 255.0.0.0

r1(config-if)#no shut

r1(config-if)#exit

r1(config)#int fa0/1

r1(config-if)#ip add 1.0.0.1 255.0.0.0

r1(config-if)#no shut

Assign IP address on router R2:

Router(config)#host R2

R2(config)#

R2(config)#int fa0/0

R2(config-if)#ip add 2.0.0.2 255.0.0.0

R2(config-if)#no shut

R2(config-if)#

R2(config-if)#exit

R2(config)#int fa0/1

R2(config-if)#ip add 3.0.0.1 255.0.0.0

R2(config-if)#no shut

Second

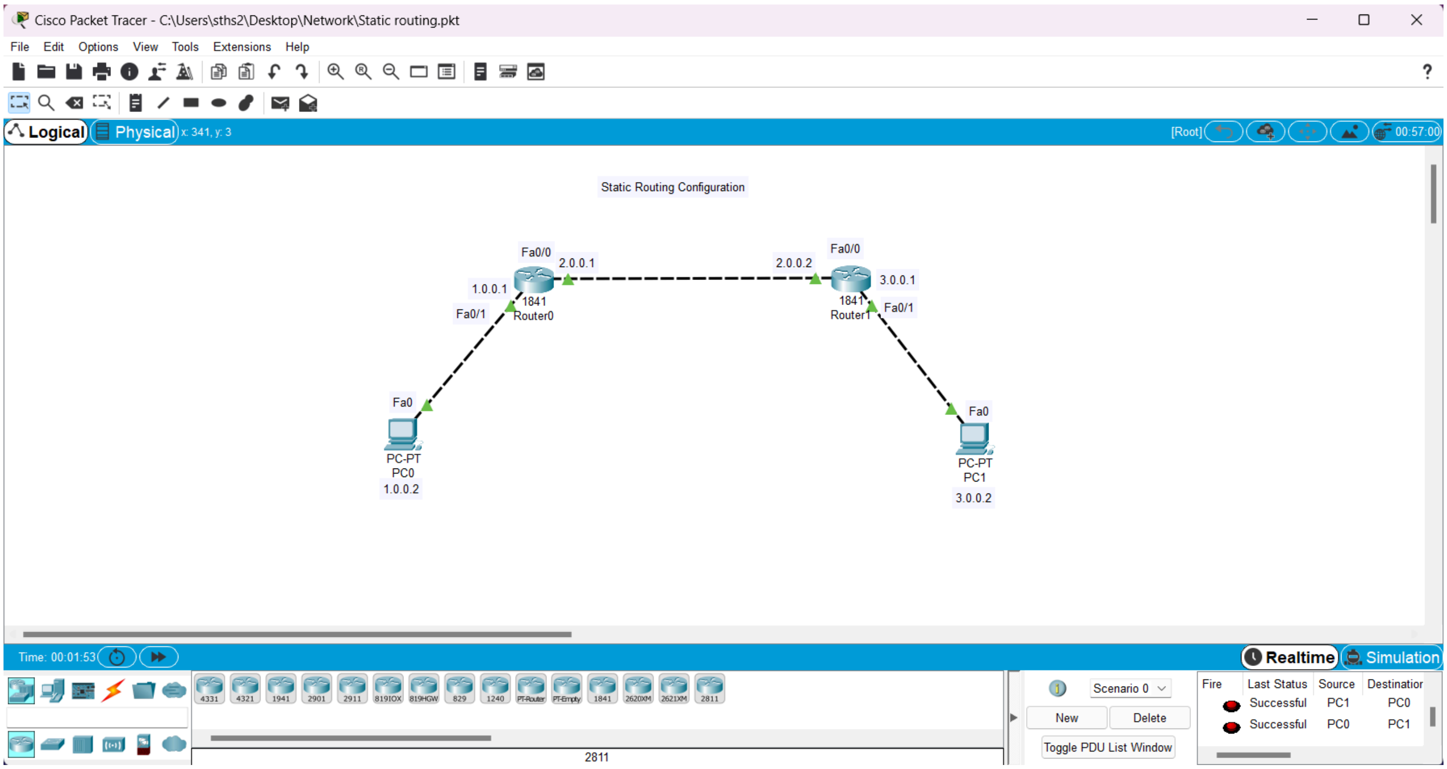
Static Routing configuration:

Static Routing configuration on Router R1:

r1(config)#ip route 3.0.0.0 255.0.0.0 2.0.0.2

Static Routing configuration on Router R2:

R2(config)#ip route 1.0.0.0 255.0.0.0 2.0.0.1

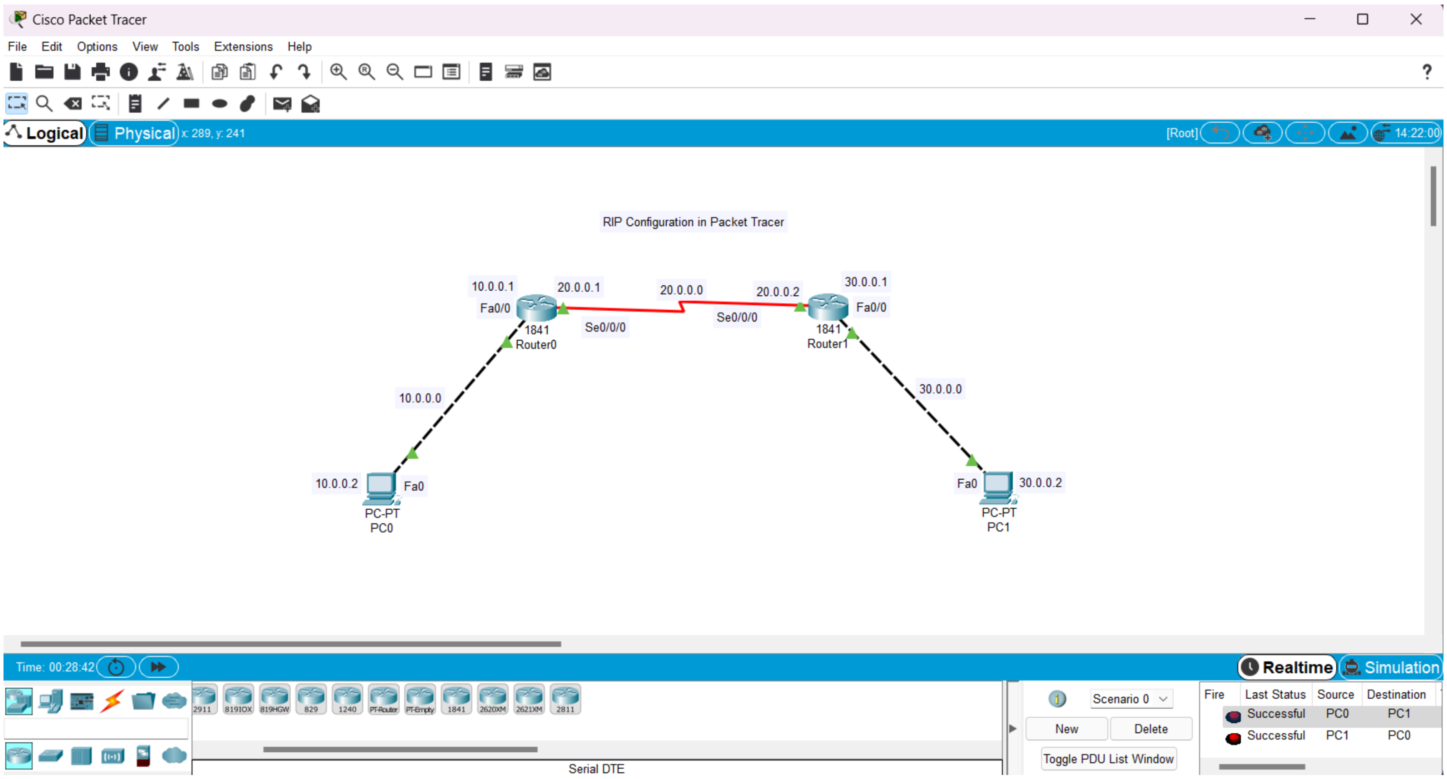


LAB 7

RIP Configuration in Packet Tracer

Configuring RIP in Packet Tracer

1)Build the network topology.



2)Configure IP addresses on the PCs and the routers.

Router 1

R1(config)#
R1(config)#int fa0/0
R1(config-if)#ip address 10.0.0.1 255.0.0.0
R1(config-if)#no shut

R1(config-if)#
R1(config-if)#int serial 0/0/0
R1(config-if)#ip add 20.0.0.1 255.0.0.0
R1(config-if)#no shut

Router 2

R2(config)#
R2(config)#int fa0/0
R2(config-if)#ip add 30.0.0.1 255.0.0.0
R2(config-if)#no shut

R2(config-if)#
R2(config-if)#int serial 0/0/0

R2(config-if)#ip add 20.0.0.2 255.0.0.0
R2(config-if)#no shut

IP configuration on PCs

Click PC->Desktop->IP Configuration. On each PC assign these addresses:

PC1: IP address: 10.0.0.2 Subnet mask 255.0.0.0 Default Gateway 10.0.0.1

PC2: IP address: 30.0.0.2 Subnet mask 255.0.0.0 Default Gateway 30.0.0.1

3)Configure **RIPv2** on the routers

Router 1

R1(config)#

R1(config)#router rip R1(config-

router)#version 2 R1(config-

router)#network 10.0.0.0 R1(config-

router)#network 20.0.0.0

Router 2

R2(config)#

R2(config)#router rip R2(config-

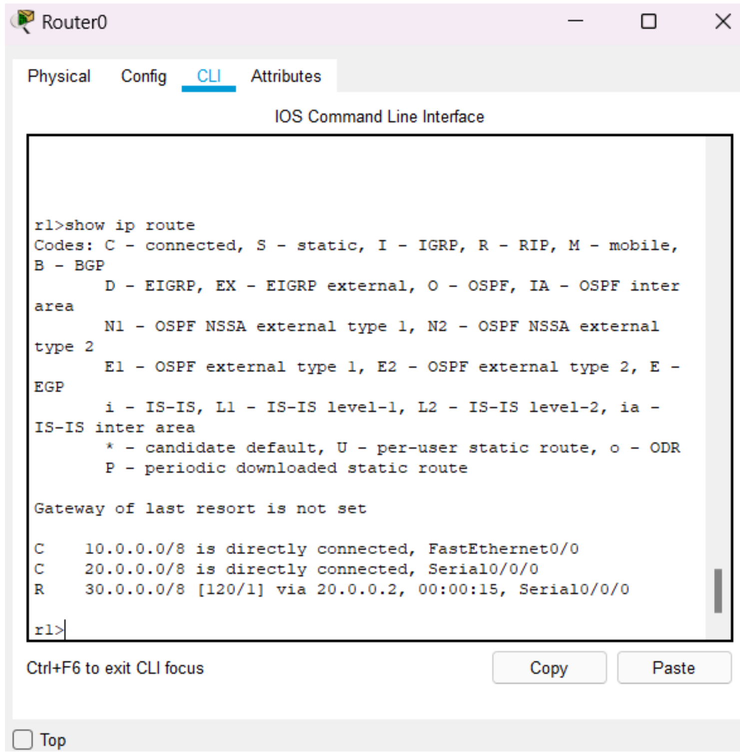
router)#version 2 R2(config-

router)#network 20.0.0.0 R2(config-

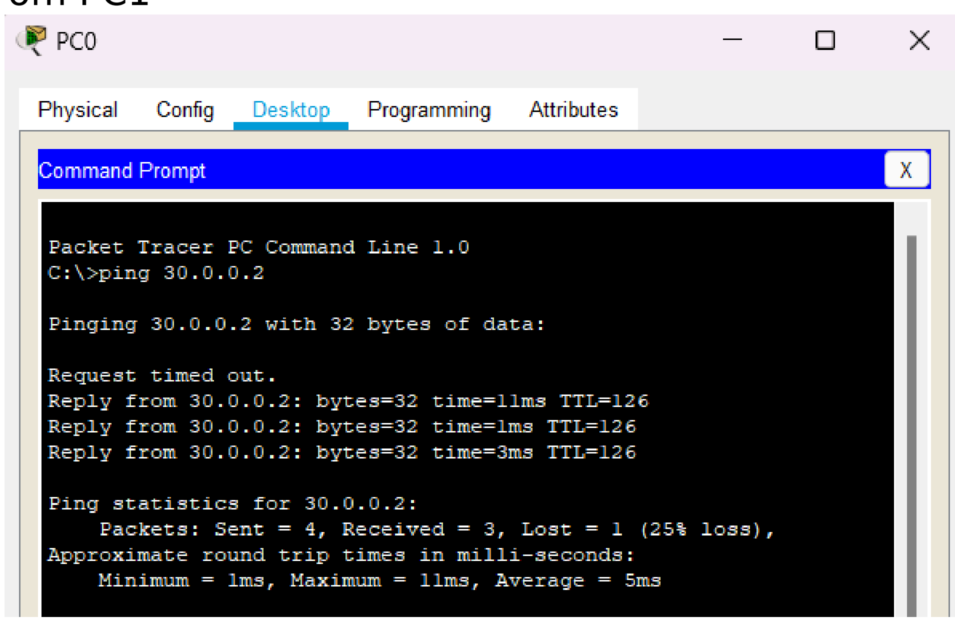
router)#network 30.0.0.0

4)Verifying RIP Configuration

To verify that RIP is in deed advertising routes, we can use the***show ip route*** command on R1.



5)Ping PC2 from PC1



LAB 8

OSPF configuration in Packet Tracer

Open Shortest Path First(OSPF) is one of the dynamic routing protocols amongst others such as EIGRP, BGP and and RIP. It is perhaps one of the most popular link state routing protocols. It is an open standard, so it can be run on routers from different vendors.

OSPF supports key features such as:

- IPv4 and IPv6 routing
- Classless routing
- Equal cost load balancing,
- Manual route summarization, etc.

OSPF has a default administrative distance of 110. It uses cost as the parameter for determining route metric. It uses the multicast address of 224.0.0.5 and 224.0.0.6 for communication between OSPF-enabled neighbors.

Routers running OSPF need to establish a neighbor relationship before exchanging routing updates. Each OSPF router runs the SFP algorithm to calculate the best routes and adds them to the routing table.

OSPF routers store routing and topology information in three tables.:

Neighbor table-which stores information about OSPF neighbors.

Topology table-stores topology structure of the network.

Routing table-stores the best routes

OSPF neighborhood discovery

Routers running OSPF need to establish a neighbor relationship before exchanging routing updates. OSPF neighbors are dynamically discovered by sending Hello packets out each OSPF-enabled interface on a router. Hello packets are sent to the multicast address of 224.0.0.5.

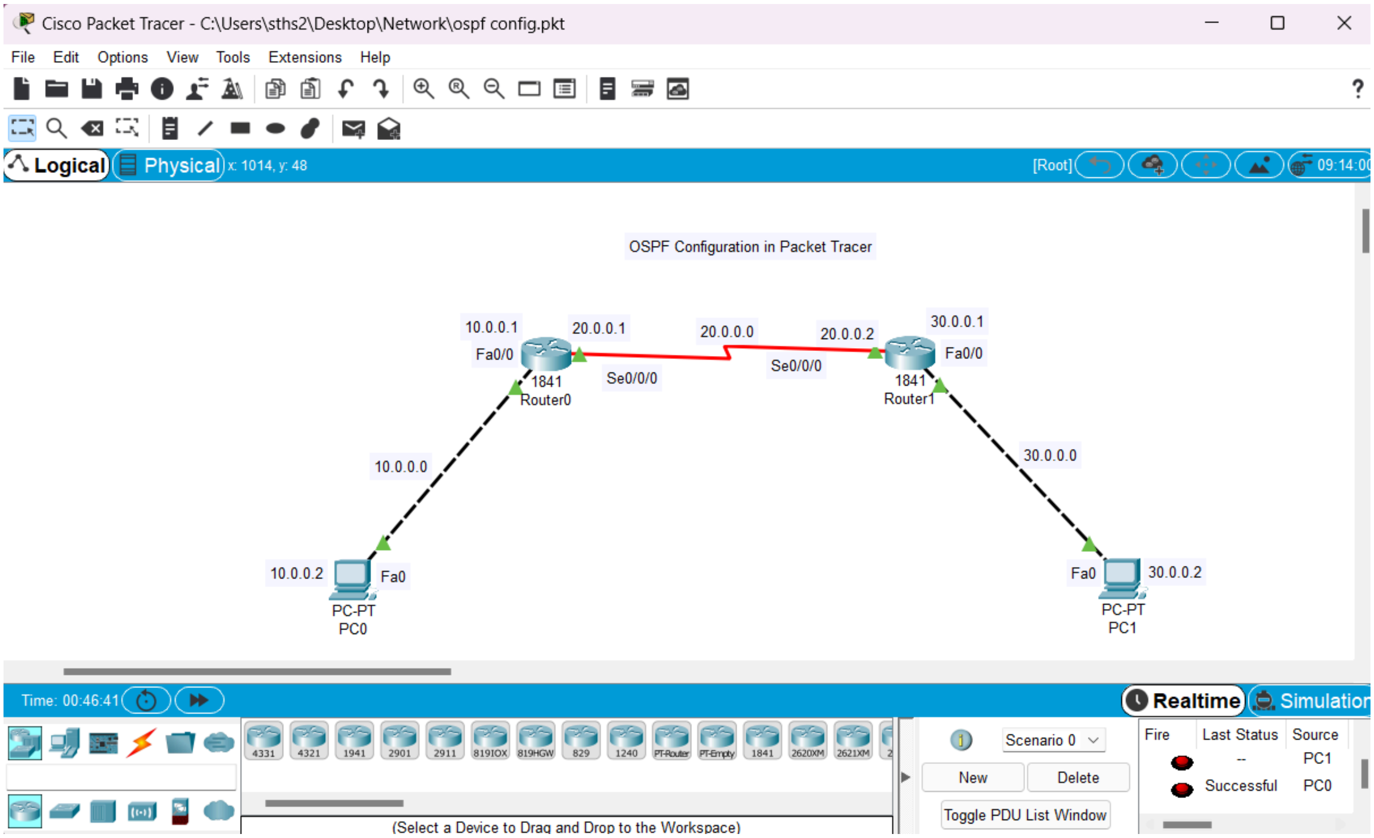
OSPF areas An area is simply a logical grouping of adjacent networks and routers. All routers in the same area have the same topology table and don't know about routers in other areas. The main benefits of using areas in an OSPF network are:

- Routing tables on the routers are reduced.
- Routing updates are reduced.
- Each area in an OSPF network must be connected to the backbone area (also known as area 0).
- All routers inside an area must have the same area ID.

A router that has interfaces in more than one area (for example area 0 and area 1) is known as an Area Border Router (ABR). A router that connects an OSPF network to other routing networks (for example, to an EIGRP network) is called an Autonomous System Border Router (ASBR).

Basic OSPF configuration

1) Build the network topology.



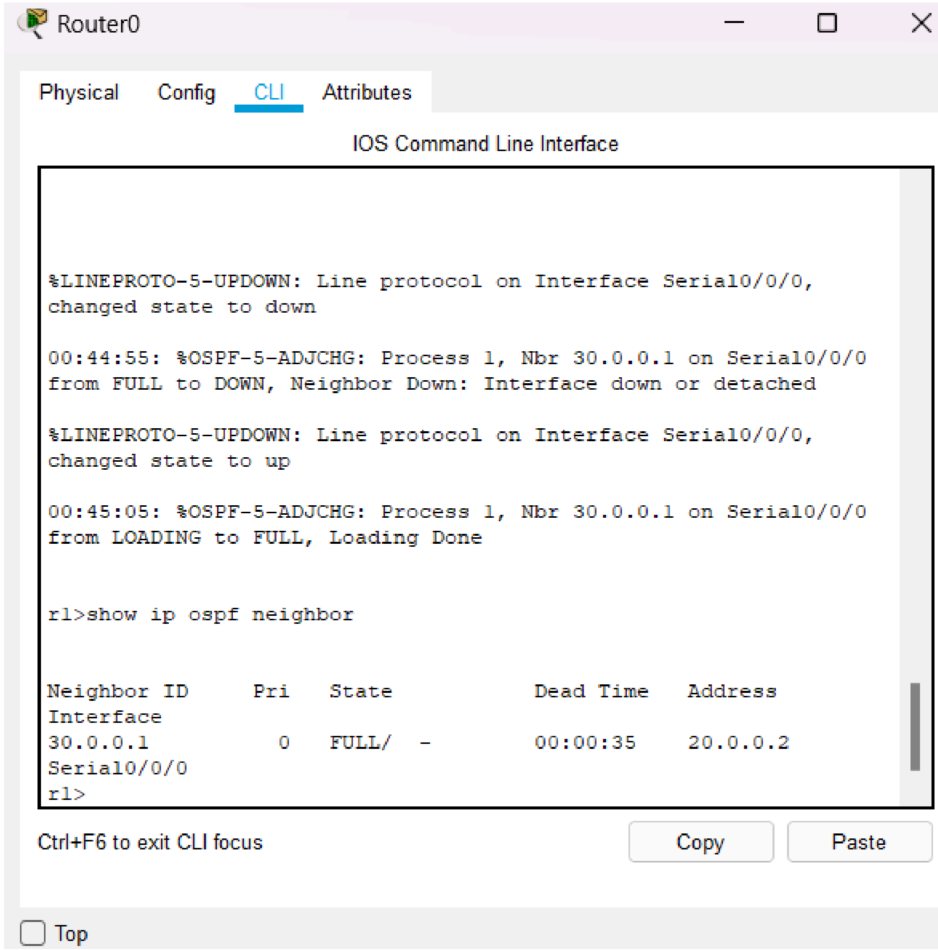
2)Configure IP addresses on PCs and router interfaces same as Lab 7.

3)Configure OSPF on the routers

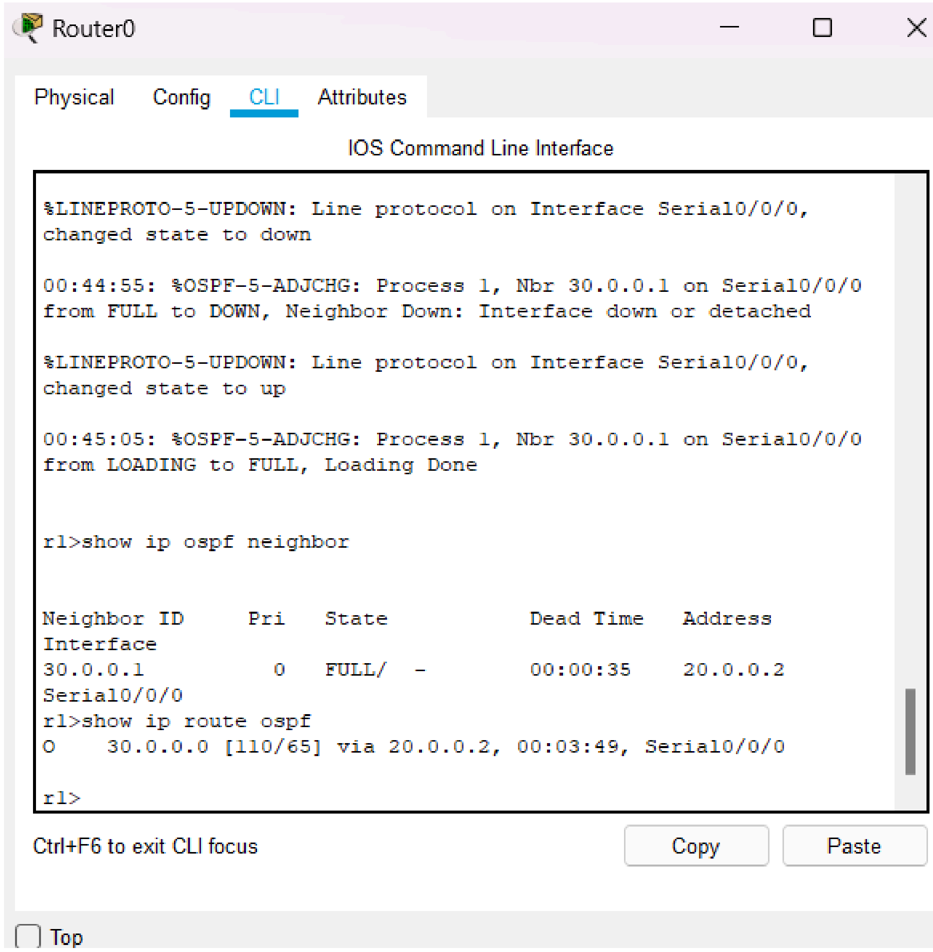
Router 1 R1(config)# R1(config)#router ospf 1 R1(config-router)#network 10.0.0.0 0.255.255.255 area 0 R1(config-router)#network 20.0.0.0 0.255.255.255 area 0 Router 2 R2(config)# R2(config)#router ospf 2 R2(config-router)#network 20.0.0.0 0.255.255.255 area 0 R2(config-router)#network 30.0.0.0 0.255.255.255 area 0

4)Verify OSPF configuration

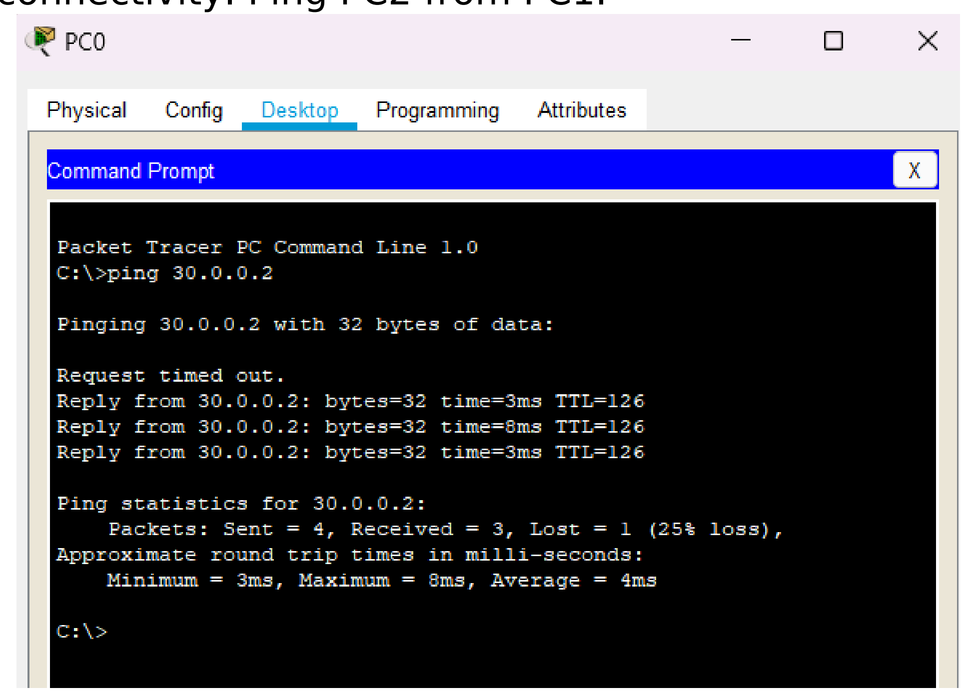
Verifying that the routers have established a neighbor relationship by typing the ***show ip ospf neighbor*** command on R1:



Verifying that R1 has learnt the route to 30.0.0.0/8 network, we’ll use ***show ip route ospf*** command on R1:



Verifying connectivity. Ping PC2 from PC1.



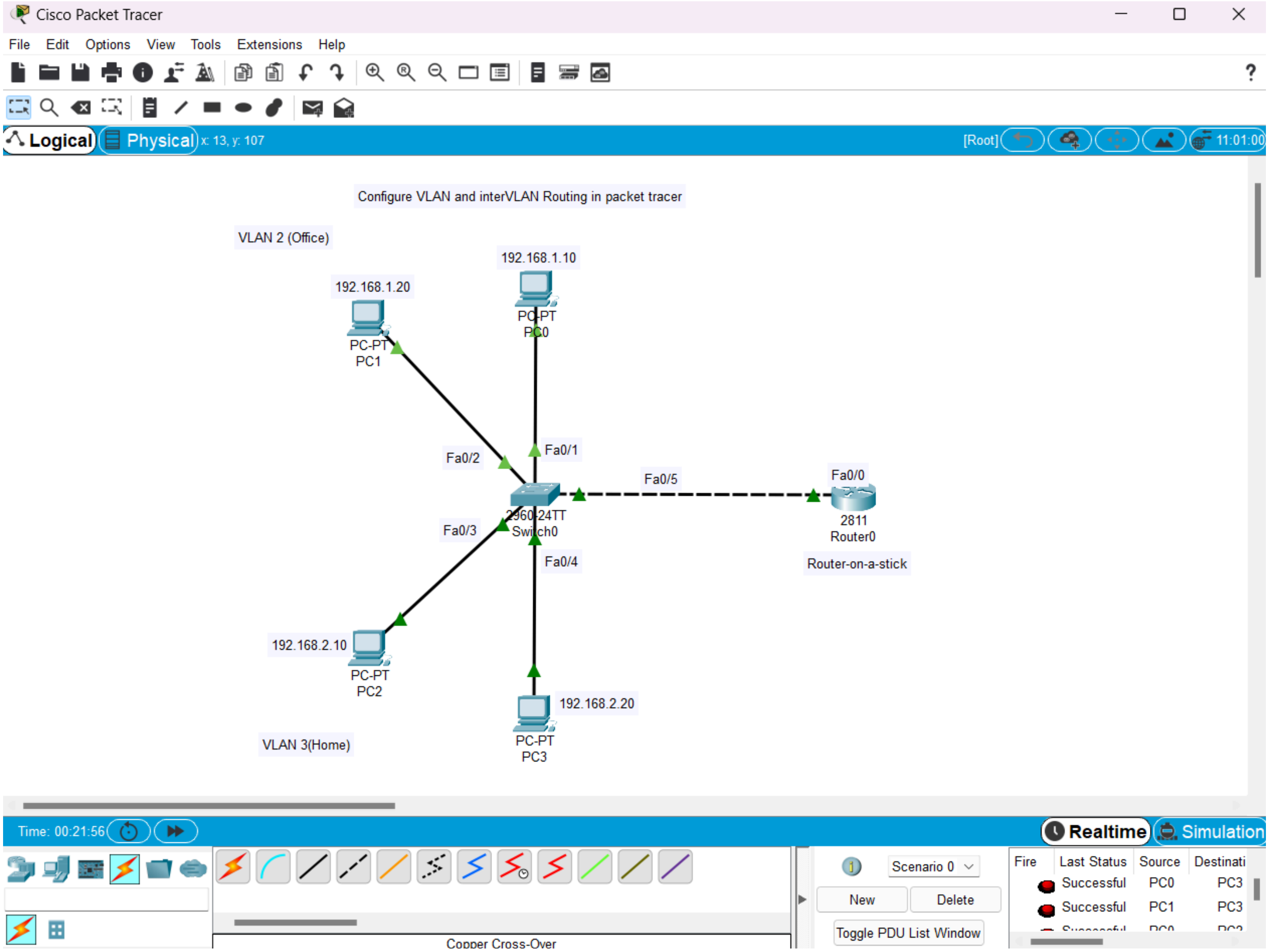
LAB 9

Configure VLAN and interVLAN Routing in Packet Tracer

A Virtual LAN (VLAN) is simply a logical LAN, just as its name suggests. VLANs have similar characteristics with those of physical LANs, only that with VLANs, you can logically group hosts even if they are physically located on separate LAN segments.

We treat each VLAN as a separate subnet or broadcast domain. For this reason, to move packets from one VLAN to another, we have to use a router or a layer 3 switch.

1) In Cisco Packet Tracer, create the network topology as shown below:



2)Create 2 VLANs on the switch: VLAN 2 and VLAN 3. You can give them custom names.

Switch#config terminal
Switch(config)#vlan 2
Switch(config-vlan)#name Office
Switch(config-vlan)#vlan 3
Switch(config-vlan)#name Home

3)Assign switch ports to the VLANs.

- An access port is assigned to a single VLAN . These ports are configured for switch ports that connect to devices with a normal network card, for example a PC in a network.

- A trunk port on the other hand is a port that can be connected to another switch or router. This port can carry traffic of multiple VLANs.

Switch>enable

Switch#config terminal

Switch(config)#int fa0/1

Switch(config-if)#switchport mode access

Switch(config-if)#switchport access vlan 2

Switch(config-if)#int fa0/2

Switch(config-if)#switchport mode access

Switch(config-if)#switchport access vlan 2

Switch(config-if)#int fa0/3

Switch(config-if)#switchport mode access

Switch(config-if)#switchport access vlan 3

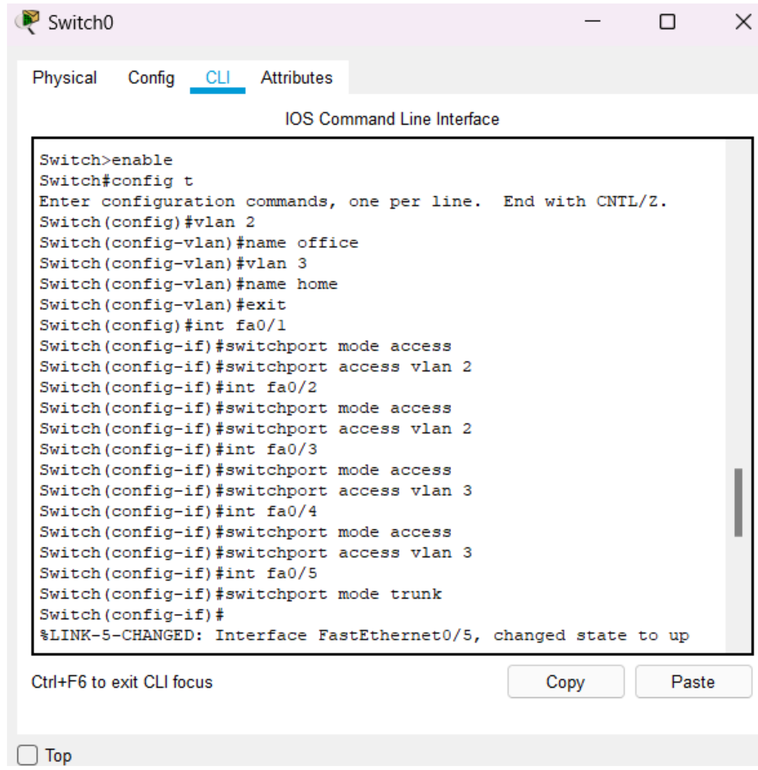
Switch(config-if)#int fa0/4

Switch(config-if)#switchport mode access

Switch(config-if)#switchport access vlan 3

Switch(config)#int fa 0/5

Switch(config-if)#switchport mode trunk



4)Assign static IP addresses to the four PCs which are located in the separate VLANs. PC0 and PC1 fall in VLAN 2 while PC2 and PC3 fall in VLAN 3.

PC1 IP address 192.168.1.10 **Subnet mask** 255.255.255.0 **Default gateway** 192.168.1.1

PC2: IP address 192.168.1.20 **Subnet mask** 255.255.255.0 **Default gateway** 192.168.1.1

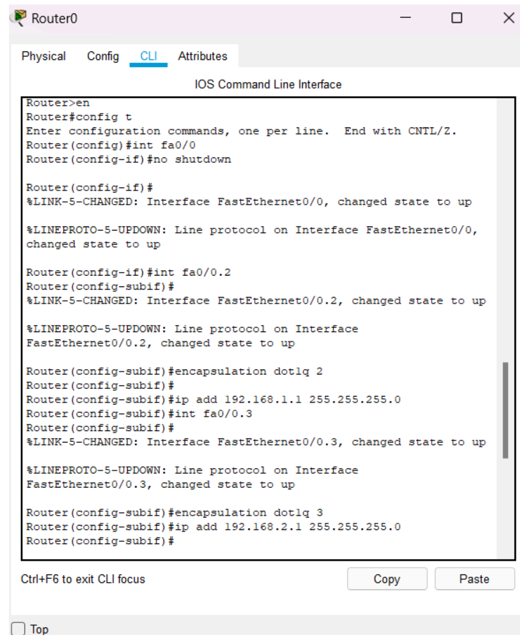
PC3: IP address 192.168.2.10 **Subnet mask** 255.255.255.0 **Default gateway** 192.168.2.1

PC4: IP address 192.168.2.20 **Subnet mask** 255.255.255.0 **Default gateway** 192.168.2.1

5)Configure inter-VLAN routing on the router

Router>enable

Router#config terminal
Router(config)#int fa0/0
Router(config-if)#no shutdown
Router(config-if)#int fa0/0.2
Router(config-subif)#encapsulation dot1q 2
Router(config-subif)#ip add 192.168.1.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#int fa0/0.3
Router(config-subif)#encapsulation dot1q 3
Router(config-subif)#ip add 192.168.2.1 255.255.255.0



6)Test inter-VLAN connectivity